ECA13 – MICROWAVE ENGINEERING: CO1 APPLICATION-BASED ASSESSMENT

Analysis, Design and Verification of a 2.45 GHz WR-340 Waveguide Feed Network for an Industrial Microwave Processing Applicator

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STAGE 1: FORMULATE THE APPLICATION AND OPERATING REQUIREMENT

1.1 Application & History: The application is an industrial microwave drying/processing line in which a 6 kW CW magnetron source feeds a batch applicator through a 1.0 m transmission link. Industrial heating matters because energy is deposited volumetrically, cutting drying time and floor space compared with convective ovens. Maxwell (1864) unified electromagnetism and Hertz (1888) demonstrated radiated waves. The 1940 cavity magnetron of Randall and Boot powered wartime radar, and in 1945 Percy Spencer at Raytheon noticed its heating effect, leading to the first microwave oven. Later milestones were satellite links, planar MMICs (1980s) and today's 5G/6G millimetre-wave front-ends.

1.2 Band Comparison: *L-band 915 MHz* (λ = 327.6 mm) penetrates deepest, but its WR-975 guide is 247.7 mm wide and impractical on a compact line. *C-band 5.8 GHz* (λ = 51.7 mm) allows a small WR-159 guide, but penetration is shallow and heating is non-uniform. *S-band 2.45 GHz ISM* (2.40–2.50 GHz, f_0 = 2.45 GHz, λ_0 = 122.4 mm, BW = 100 MHz) gives a good penetration depth in water-bearing loads, negligible atmospheric loss, low-cost mass-produced magnetrons and a licence-free ISM allocation, so it is selected.

1.3 Constraints: Leakage must stay ≤ 5 mW/cm² at 5 cm (oven-type limit, 21 CFR 1030.10, used as the design target), so a fully enclosed metal structure with choked joints is needed. Reflected power must be limited to protect the magnetron. Factory ambient is 0–60 °C with dust and humidity, and 6 kW CW heats the walls, so a robust, easily cleaned guide is needed. These constraints rule out open structures such as microstrip.

Table 1 — Quantified Requirement Specification

Parameter	Quantified Requirement	Engineering Justification
Application / Category	Industrial microwave processing feed, 1.0 m link	Magnetron (6 kW CW) → applicator
Frequency / BW	2.40–2.50 GHz (f_0 = 2.45 GHz)	ISM band; λ_0 = 122.4 mm
Impedance	Source 50 Ω probe; $Z_{TE10} \approx 534$ Ω	Matched through coax-to-waveguide launcher
Max insertion loss	IL ≤ 0.10 dB over 1.0 m	Keeps >97.7% of power; limits wall heating
Return loss / VSWR	RL ≥ 22 dB; VSWR ≤ 1.17:1	Protects magnetron from reflected power
Isolation / EMI	SE ≥ 80 dB; leakage ≤ 5 mW/cm² at 5 cm	Operator safety, ISM emission limits
Power, size, weight, cost	6 kW CW; ≤ 90×50 mm; ≤ 3 kg/m; ≤ US$300/m	Plant-floor installation and budget

STAGE 2: EVALUATE AND SELECT THE TRANSMISSION STRUCTURE

2.1 Modes: Coax carries pure TEM (no cutoff); its first higher mode is TE_{11}, near 6.0 GHz for a 7/8-in line. Stripline is TEM in a homogeneous dielectric. Microstrip is quasi-TEM, because the air–substrate interface makes the phase velocities of the E_z/H_z components differ, giving ε_{eff} between 1 and ε_r and dispersion. A hollow waveguide cannot support TEM: with one boundary conductor, $\nabla_t^2\Phi = 0$ and Φ = const give $E_t = 0$, and Ampère's law needs an enclosed return conductor. Waveguides carry TE and TM modes only, with TE_{10} dominant (the lowest TM mode is TM_{11}).

Table 2 — Weighted Decision Matrix (Score / 5)

Criterion (Weight)	Coax (7/8-in)	Waveguide (WR-340)	Stripline	Microstrip
Insertion loss (0.25)	4 (≈0.04 dB/m)	5 (≈0.016 dB/m)	2 (≈0.45 dB/m)	1 (≈0.80 dB/m)
Power handling (0.25)	2 (~1.4 kW)	5 (>100 kW)	1 (<0.3 kW)	1 (<0.2 kW)
Shielding (0.15)	5 (solid outer)	5 (enclosed walls)	4 (via fence)	1 (open)
Single mode (0.10)	4 (TEM pure)	5 (strict TE_{10})	4 (plate modes)	2 (quasi-TEM)
Size & weight (0.10)	4 (22 mm OD)	2 (86.4×43.2 mm)	4 (planar)	5 (compact)
Cost (0.08)	3	4 (Al extrusion)	3	5
Integration (0.07)	3 (needs adapters)	5 (magnetron flange)	2	3
WEIGHTED TOTAL	**3.50**	**4.62 (✓ Selected)**	**2.53**	**1.96**

Sensitivity & defence: The strongest alternative is coax. Raising the size/weight weight (others scaled proportionally) from 0.10 reverses the ranking at about 0.42, where coax (≈3.7) overtakes waveguide. This is unlikely because power handling is a hard requirement: 6 kW CW exceeds the ≈1.4 kW rating of 7/8-in coax, which would need a much larger line and still have breakdown risk.

► Design Decision 1: I select **WR-340 rectangular waveguide (TE_{10})** over **7/8-in coax** because it handles >100 kW against ≈1.4 kW, has ≈0.016 dB/m loss against ≈0.04 dB/m, and is fully shielded. **Frequency sensitivity:** halving f_0 to ≈1.2 GHz is below the 1.736 GHz cutoff (evanescent), so WR-975 would be needed. Doubling to ≈4.9 GHz makes WR-340 overmoded (TE_{20}/TE_{01} at 3.47 GHz), so WR-187 would be needed. At higher frequency, coax loss rises as $\sqrt{f}$ and its power rating falls further, so waveguide gains more.

STAGE 3: DEVELOP THE MODE AND PROPAGATION MODEL

3.1 Telegrapher's model: $-dV/dz = (R + j\omega L)I$ and $-dI/dz = (G + j\omega C)V$ give $d^2V/dz^2 = \gamma^2 V$ with $\gamma = \alpha + j\beta = \sqrt{[(R + j\omega L)(G + j\omega C)]}$, $Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$. Low loss: $\alpha \approx R/(2Z_0) + GZ_0/2$, $\beta \approx \omega\sqrt{(LC)}$, $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$.

3.2 Field derivation: Assuming a source-free, lossless air-filled guide with PEC walls and $e^{j\omega t - \gamma z}$ dependence, Maxwell's equations reduce for TE modes to $(\partial^2/\partial x^2 + \partial^2/\partial y^2 + k_c^2)H_z = 0$. The condition $\partial H_z/\partial n = 0$ gives $k_{c,mn} = \sqrt{[(m\pi/a)^2 + (n\pi/b)^2]}$ and $f_{c,mn} = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$. For TE_{10}: $H_z = H_0\cos(\pi x/a)e^{-j\beta z}$, $E_y = -(j\omega\mu a/\pi)H_0\sin(\pi x/a)e^{-j\beta z}$. **Mode comparison:** TE modes have $E_z = 0$ and TM modes $H_z = 0$. Both are dispersive, with $Z_{TE} = \eta/\xi$ (rising near cutoff) and $Z_{TM} = \eta\xi$. TE_{10} has the lowest cutoff, the widest single-mode band and the simplest field, so it is best.

Table 3 — Designed dimensions and propagation parameters at f_0 = 2.45 GHz

Quantity	Value	Quantity	Value
a × b (WR-340)	86.36 × 43.18 mm	$\beta = \sqrt{(k_0^2 - k_c^2)}$	$\sqrt{(51.35^2 - 36.38^2)} = 36.24$ rad/m
$f_{c,10} = c/2a$	1.736 GHz	$\lambda_g = 2\pi/\beta$	173.4 mm
$f_{c,20} = f_{c,01}$ (degenerate)	3.472 GHz	$\xi = \sqrt{(1 - (f_c/f)^2)}$	0.7057
$f_{c,11}$ (TE/TM)	3.881 GHz	$v_p = c/\xi$; $v_g = c\xi$	4.248×10^8 ; 2.116×10^8 m/s
Single-mode range	1.736–3.472 GHz	Group delay $\tau_g = L/v_g$	4.726 ns (1.0 m)
Margin at 2.40 / 2.50 GHz	38.3% / 38.9%	$Z_{TE10} = \eta_0/\xi$	376.73/0.7057 = 533.8 Ω

Worked calculation: $f_c = (2.998\times10^8\ \text{m/s})/(2 \times 0.08636\ \text{m}) = 1.736$ GHz; $\lambda_0 = c/f_0 = 122.4$ mm; $k_0 = 2\pi/\lambda_0 = 51.35$ rad/m; $k_c = \pi/a = 36.38$ rad/m; $\beta = 36.24$ rad/m; $\lambda_g = 173.4$ mm.

► Design Decision 2: The dominant mode is **TE_{10}** with cutoff **1.736 GHz**; the single-mode range is **1.736** to **3.472 GHz**; the dimensions are **acceptable** because 2.40–2.50 GHz has ≈38% margin on both sides. **Higher-order risk:** since a = 2b, TE_{20} and TE_{01} are degenerate at 3.472 GHz. The launcher probe (E parallel to b) is centred at x = a/2, which nulls TE_{20} and does not excite TE_{01} (E_x). Magnetron harmonics (4.90 GHz, 7.35 GHz) would propagate in higher modes, so a harmonic low-pass filter or choke is added at the launcher.

STAGE 4: EVALUATE LOSSES AND DESIGN A LOSS-REDUCTION STRATEGY

Worked calculation (initial, Al-6061, $\sigma = 2.5\times10^7$ S/m, $R_q = 2.0$ µm): $\delta_s = 1/\sqrt{(\pi f \mu_0 \sigma)} = 2.034$ µm; $R_{s0} = 1/(\sigma\delta_s) = 0.0197$ Ω/sq; $K_{sr} = 1 + (2/\pi)\arctan[1.4(R_q/\delta_s)^2] = 1.595$; $R_{s,eff} = 0.0314$ Ω/sq. $\alpha_c = [R_{s,eff}/(b\eta_0\xi)][1 + (2b/a)(f_c/f)^2] = 0.0357$ dB/m. Dielectric (nylon-type window, $\varepsilon_r = 3.0$, $\tan\delta = 1.5\times10^{-3}$, 6 mm): $\alpha_d \approx 27.3\cdot\tan\delta\cdot\sqrt{\varepsilon_r}\cdot(t/\lambda_0) = 0.004$ dB. **Redesign (10 µm Cu plate, $\sigma = 5.8\times10^7$ S/m, $R_q = 0.4$ µm):** $\delta_s = 1.335$ µm (plating $7.5\delta_s$), $K_{sr} = 1.080$, $R_{s,eff} = 0.0139$ Ω/sq, $\alpha_c = 0.0158$ dB/m. PTFE window ($\tan\delta = 2\times10^{-4}$): 0.001 dB.

Table 4 — Loss Budget (dB over L = 1.0 m, i.e. dB/m)

Loss component	Initial	% of total	Improved	Trade-off / Remarks
Conductor	0.036	31.9%	0.016	Cu plating + electropolish: cost +12%
Dielectric (window)	0.004	3.5%	0.001	PTFE window is softer, so it needs support
Radiation (flange leakage)	0.002	1.8%	0.001	Choke flanges + conductive gaskets
Connector / transition	0.020	17.7%	0.010	Tuned launcher with harmonic filter
Mismatch (|Γ| = 0.108 → 0.025)	0.051	45.1%	0.003	QWT adds 43 mm length
Aging margin (10 yr)	—	—	0.003	Cu oxide / tarnish allowance
Total attenuation	**0.113 (FAIL)**	**100%**	**0.034 (PASS)**	**≤ 0.10 dB met, margin 0.066 dB**

Parametric study (α_c at 2.45 GHz, dB/m): Al/R_q = 2.0 µm: 0.0357; Al/0.4 µm: 0.0231; Cu/2.0 µm: 0.0265; Cu/0.4 µm: 0.0158; Ag/0.4 µm: 0.0154. Across 2.40–2.50 GHz α_c falls slightly (0.0364 → 0.0350 for Al) because the guide moves away from cutoff. Smoothing the surface and raising σ both help, and roughness matters most while R_q is comparable to δ_s. Silver gives only 2.5% gain over Cu, so it is not justified. Dielectric loss tangent only matters in the window.

► Design Decision 3: The dominant loss mechanism is **mismatch (reflection)**, contributing **45.1%** of the total (conductor is second at 31.9%). I adopt **a quarter-wave transformer plus 10 µm Cu plating with electropolish**, reducing attenuation from **0.113 to 0.034 dB**. The improved design meets IL ≤ 0.10 dB; the trade-off is +12% cost and 43 mm extra length.

STAGE 5: NETWORK MODEL, MATCHING & S-PARAMETERS

5.1 Equivalent V/I: In non-TEM modes $V = \int E\cdot dl$ depends on the path and I on the contour, so neither is unique. The power-voltage (P-V) convention defines $V = E_{y,max}\cdot b$ and $I = 2P/V^*$ (P = average power), giving $Z_{0,PV} = (2b/a)\cdot Z_{TE10} = (2 \times 43.18/86.36) \times 533.8 =$ **533.8 Ω** (since a = 2b). **Wave variables:** $a_n = (V + Z_0 I)/(2\sqrt{Z_0})$, $b_n = (V - Z_0 I)/(2\sqrt{Z_0})$, $S_{11} = b_1/a_1|_{a2=0}$.
5.2 Load mismatch: Applicator $Z_L = 430$ Ω. $|\Gamma| = |430 - 533.8|/|430 + 533.8| = 0.1077$; RL = 19.36 dB; VSWR = 1.241:1 — **FAILS RL ≥ 22 dB**.
5.3 Quarter-wave transformer: $Z_T = \sqrt{(Z_{0,PV}\cdot Z_L)} = \sqrt{(533.8 \times 430)} = 479.1$ Ω; $l_q = \lambda_g/4 = 173.4/4 = 43.35$ mm; $b_T = b\cdot Z_T/Z_{0,PV} = 43.18 \times 479.1/533.8 = 38.75$ mm (a unchanged). **5.4 ABCD → S:** each section is [[cosθ, jZ sinθ],[j sinθ/Z, cosθ]], θ = βl. Cascading the 1.0 m line and the QWT, $S_{11} = (A + B/Z_0 - CZ_0 - D)/(A + B/Z_0 + CZ_0 + D)$ and $S_{21} = 2/(A + B/Z_0 + CZ_0 + D)$. Ideal $|S_{11}| = 0$ at f_0 and 0.013 (RL 37.5 dB) at 2.40/2.50 GHz. With step-fringing and machining allowance the budget is $|S_{11}| \le 0.025$ (RL ≥ 32.0 dB, VSWR 1.051). $|S_{21}| = 0.996$ (IL = 0.034 dB), $S_{12} = S_{21}$ (reciprocal) and $|S_{11}|^2 + |S_{21}|^2 = 0.993$ (0.7% dissipated). A NumPy ABCD cascade confirms these values (appendix).

Table 5 — Network Summary

Parameter	Before Matching	After Matching	Target	Interpretation
$Z_{0,PV}$ / Z_L	533.8 / 430 Ω	Z_T = 479.1 Ω step	50 Ω port	Step bridges load to guide
|Γ| / VSWR	0.1077 / 1.241:1	≤ 0.025 / 1.051:1	≤ 1.17:1	Reflected power 1.16% → 0.06%
Return loss (S_{11})	19.36 dB (FAIL)	≥ 32.0 dB	≥ 22 dB	12.7 dB improvement
Insertion loss (S_{21})	0.113 dB (Al)	0.034 dB (Cu + QWT)	≤ 0.10 dB	0.079 dB saved (+1.8% power)
Reciprocity / lossless	$S_{12} = S_{21}$; Σ = 0.986	$S_{12} = S_{21}$; Σ = 0.993	Σ ≥ 0.98	Passive, symmetric, nearly lossless

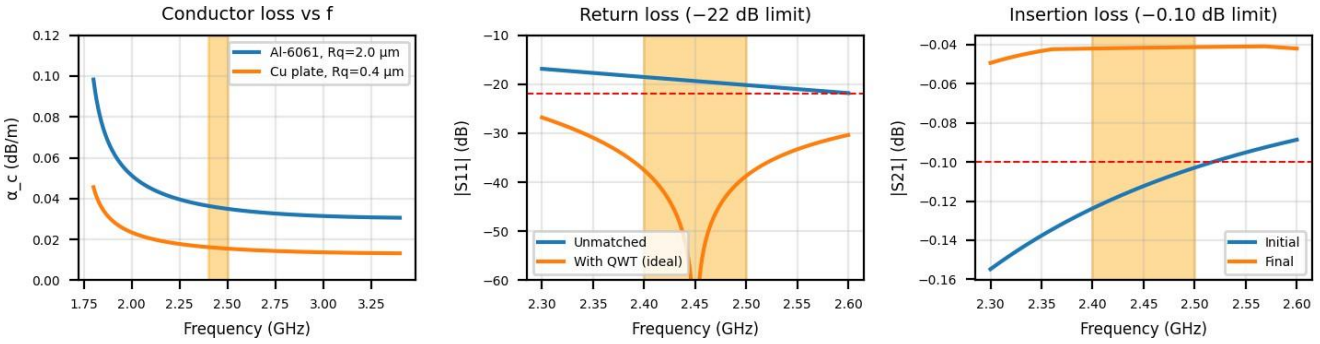


Figure 1 — (left) conductor attenuation vs frequency, (centre) |S11| with and without the QWT, (right) |S21|; the ISM band 2.40–2.50 GHz is shaded and the limits are dashed.

STAGE 6: VALIDATE, CRITIQUE AND DEFEND THE FINAL DESIGN

6.1 Validation & critique: Cutoff and impedance were cross-checked against the standard WR-340 data (1.736 GHz) and the Python cascade. The model assumes ideal uniform air fill, neglects the launcher and window reactance and treats the step as lossless, so real RL is a few dB lower, which the 0.025 residual reflection budget covers. **Tolerance:** a ±5% error on a (a′ = 82.04 mm) shifts f_c to 1.827 GHz (still single-mode) and $Z_{0,PV}$ to 595 Ω, giving ideal RL ≈ 25.0 dB and ≈ 21.8 dB with the residue added in phase, which is marginally below 22 dB. With CNC tolerance ±0.1 mm (±0.12%) the worst-case RL is ≥ 31 dB.
6.2 EMC / safety and test plan: Choke flanges, conductive gaskets and bonded grounding hold SE above 80 dB. Tests: (1) VNA 2–3 GHz with TRL waveguide calibration for S_{11}, S_{21} and delay; (2) leakage survey with a calibrated meter at 5 cm around flanges at 6 kW (≤ 5 mW/cm²); (3) thermal rise on walls after 1 h at 6 kW; (4) harmonic emission scan to 8 GHz. **Improvement:** add a ferrite circulator with a

water dummy load to absorb residual reflection.

Table 6 — Specification Verification

Requirement	Specified	Initial	Final (worst case)	Satisfied?
Frequency / single-mode range	2.40–2.50 GHz	1.736–3.472 GHz	1.74–3.47 GHz (±0.1 mm)	YES
Dominant mode / cutoff	TE_{10}, >30% margin	1.736 GHz, 38%	1.734–1.738 GHz, 38%	YES
Insertion loss	≤ 0.10 dB/m	0.113 dB (FAIL)	0.034 dB (incl. aging)	YES
Impedance / VSWR	≤ 1.17:1	1.241:1 (FAIL)	≤ 1.051:1	YES
Return loss	≥ 22 dB	19.36 dB (FAIL)	≥ 31 dB (±0.1 mm)	YES
EMI shielding	≥ 80 dB	>80 dB (enclosed)	>100 dB + choke	YES
Power / size	6 kW; ≤ 90×50 mm	>100 kW; 86.4×43.2 mm	unchanged; 43 mm longer	YES

FINAL ENGINEERING DECISION

(i) Initial design (Al-6061, unmatched): NO. IL = 0.113 dB exceeds 0.10 dB and RL = 19.36 dB fails ≥ 22 dB; the dominant loss is mismatch (45.1%). **(ii) Final design (Cu-plated WR-340 + QWT): YES.** The electropolished Cu-plated WR-340 with Z_T = 479.1 Ω, l_q = 43.35 mm and b_T = 38.75 mm gives IL = 0.034 dB, RL ≥ 32 dB, VSWR ≤ 1.051:1, SE > 100 dB and a 38% single-mode margin at 6 kW. The only sensitivity is the broad-wall tolerance, so a ±5% error would marginally breach RL; ±0.1 mm CNC avoids this. The design is accepted for manufacture.

Appendix — Python verification (NumPy)

```
import numpy as np
c=2.998e8; a,b=86.36e-3,43.18e-3; eta=376.73; ZL=430.0
fc=c/(2*a); Zte=lambda f: eta/np.sqrt(1-(fc/f)**2); lg=lambda f: c/f/np.sqrt(1-(fc/f)**2) f0=2.45e9; Z0=Zte(f0);
ZT=np.sqrt(Z0*ZL); lq=lg(f0)/4
def gamma_in(f):
    th=2*np.pi/lg(f)*lq; Zin=ZT*(ZL+1j*ZT*np.tan(th))/(ZT+1j*ZL*np.tan(th)) return abs((Zin-
    Zte(f))/(Zin+Zte(f)))
print(gamma_in(2.40e9), gamma_in(2.50e9))   # -> 0.0133, 0.0116
```

References

[1] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2012. [2] R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed., IEEE Press/Wiley, 2001. [3] Ramo, Whinnery, Van Duzer, *Fields and Waves in Communication Electronics*, 3rd ed., Wiley, 1994. [4] N. Marcuvitz, *Waveguide Handbook*, McGraw-Hill, 1951.